

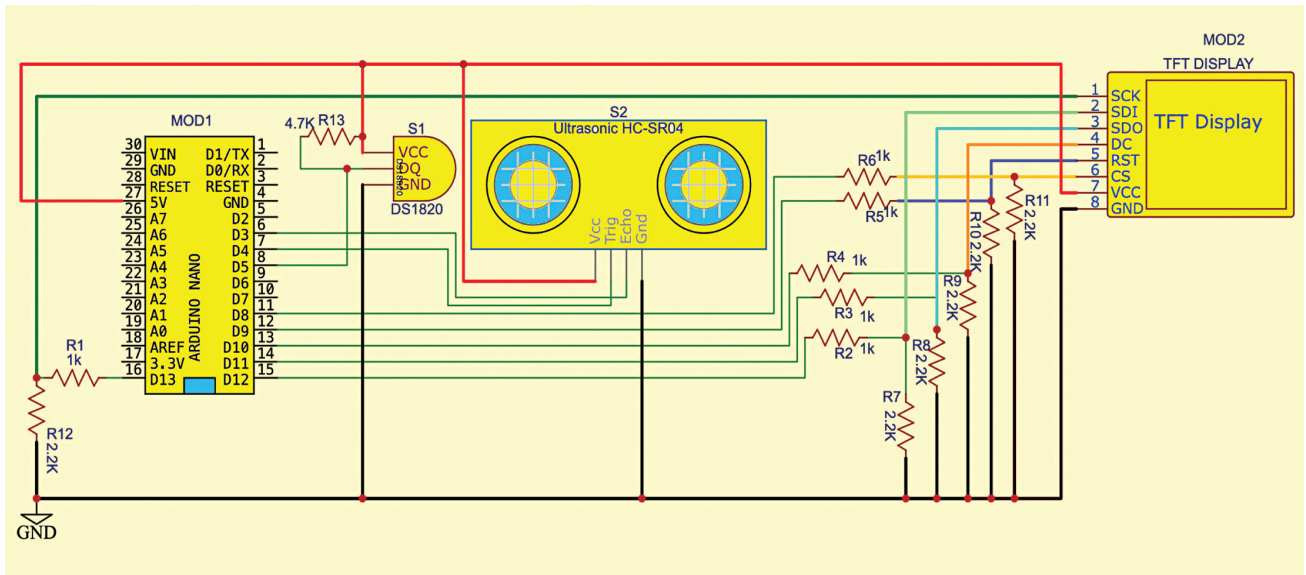


Fig. 3: Circuit diagram

```

1  #include <LCDWIKI_GUI.h> //Core graphics library
2  #include <LCDWIKI_SPI.h> //Hardware-specific library
3  #include <OneWire.h>
4  #include <DallasTemperature.h>
5
6  #define MODEL ILI9488_18
7  #define CS 8
8  #define CD 10
9  #define RST 9
10 #define LED -1
11 //define some colour values
12 #define BLACK 0x0000
13 #define BLUE 0x001F
14 #define RED 0xF800
15 #define ORANGE 0xFBE7
16 #define GREEN 0x07E0
17 #define LGREEN 0x3FE7
18 #define CYAN 0x07FF
19 #define MAGENTA 0xF81F
20 #define YELLOW 0xFFE0
21 #define LPINK 0xE0FE
22 #define LEVNDR 0x9C7F
23 #define PINK 0xE0F0
24 #define ONE_WIRE_BUS 4
25
26 LCDWIKI_SPI mylcd(MODEL,CS,CD,RST,LED);
27 OneWire oneWire(ONE_WIRE_BUS);
28 DallasTemperature sensors(&oneWire);
29 const int trig = 2,echo=3;
30 long duration;
31 float water_level_in_cm,water_temp;
32 int water_level_in_ltr,water_level_prcntg,water_tem;
33 void TFT_LCD_init()

```

Fig. 4: Code snippet

level and temperature monitoring on a 10.92cm (4.3-inch) TFT LCD using graphics and colour-changing animation of water filling a tank, along with a rising temperature bar graph. Fig. 1 shows the prototype on the left and a snapshot of the

TFT display on the right. The components required to build the system are listed in Table 1.

Circuit and working

Fig. 2 shows the hardware connections, and Fig. 3 shows the circuit diagram. The system is built around the HC-SR04 ultrasonic sensor (S2), DS18B20 waterproof temperature sensor (S1), Arduino Nano (MOD1), a 10.92cm (4.3-inch) SPI TFT display (MOD2), and a few other components. The TFT display (480 × 320 resolution) shows the actual water level (in litres) and temperature. Resistors are used for signal

conditioning, pull-up, and voltage-level shifting.

The TRIG and ECHO pins of the ultrasonic sensor connect to D4 and D3, respectively, to measure distance by calculating the time between pulse transmission and

echo reception. The TFT display connects through SPI lines: SCK to D13, SDI to D12, SDO to D11, DC to D10, RST to D9, and CS to D8. Resistors R1 through R12 ensure the safe operation of the TFT module.

The Arduino is powered via the 5V rail, with a common ground across all modules. The HC-SR04 measures the distance to the water surface to compute the level, while the DS18B20 provides temperature data. The Arduino processes both and displays them graphically, using tank animations and bar graphs. The resistors protect components during operation.

The temperature sensor probe DS18B20 has three interfacing pins: + V, op, and GND. The + V pin is connected to the 5V output from the Arduino board. GND pin is connected to the circuit ground. DQ pin is connected to digital pin 5 (D5). One 4.7k pull-up is connected between + V and DQ pin as it is recommended in the datasheet for the DS18B20 sensor. The TFT LCD module has nine interfacing pins, which are connected to the Arduino Nano board, as shown in Table 2.

There is a voltage divider network (built using 1kΩ and 2.2kΩ resistors) between some of the Arduino digital pins and the